# Supplementary Material

*Comparative Effectiveness and Safety of Apixaban Versus Rivaroxaban for Stroke Prevention in Atrial Fibrillation Patients with Chronic Kidney Disease Stage 3b–5: A Target Trial Emulation Using Electronic Health Record Data*

*April 2026*

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## Supplementary Figure S1

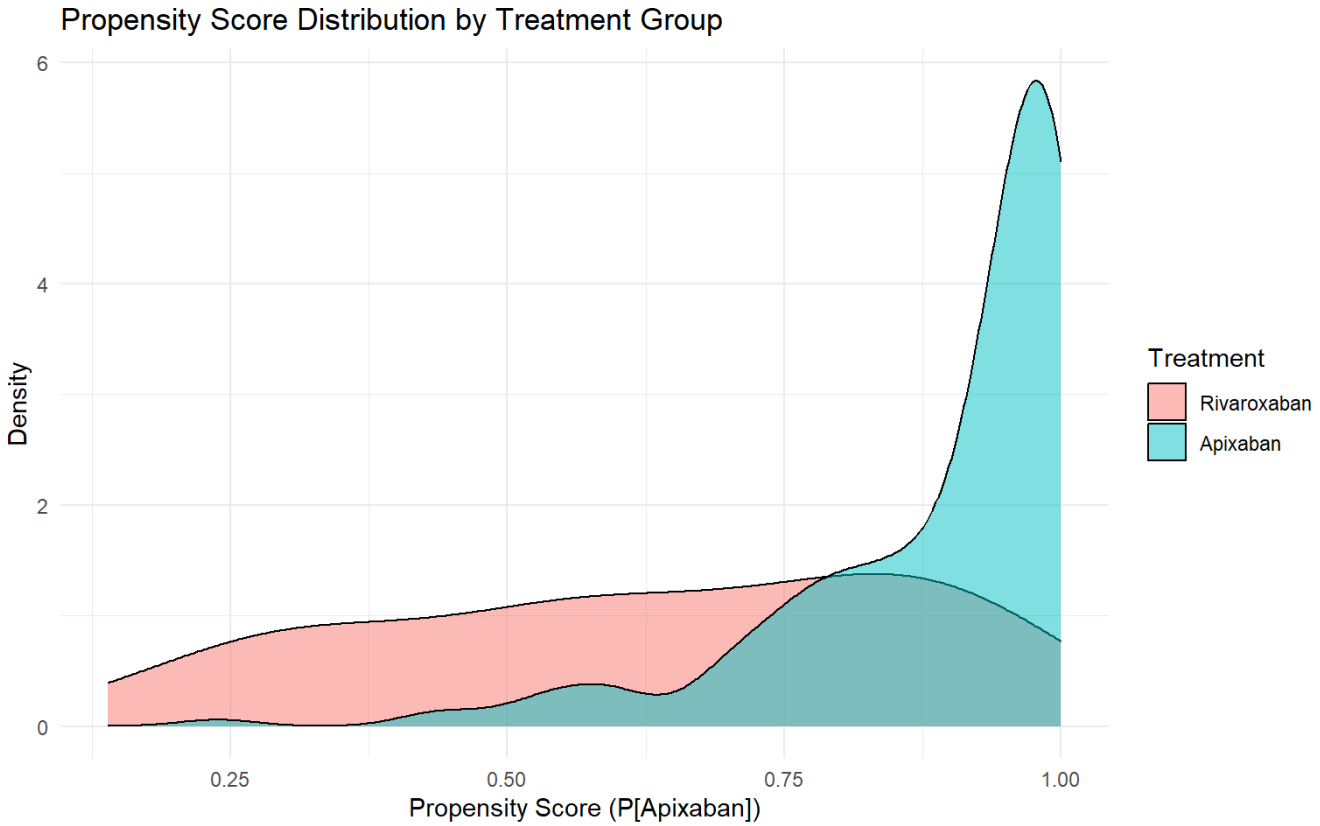



***Figure S1.*** *Propensity score distributions by treatment group. The overlap between apixaban (blue) and rivaroxaban (red) distributions indicates the region of common support for inverse probability weighting.*

## Supplementary Figure S2

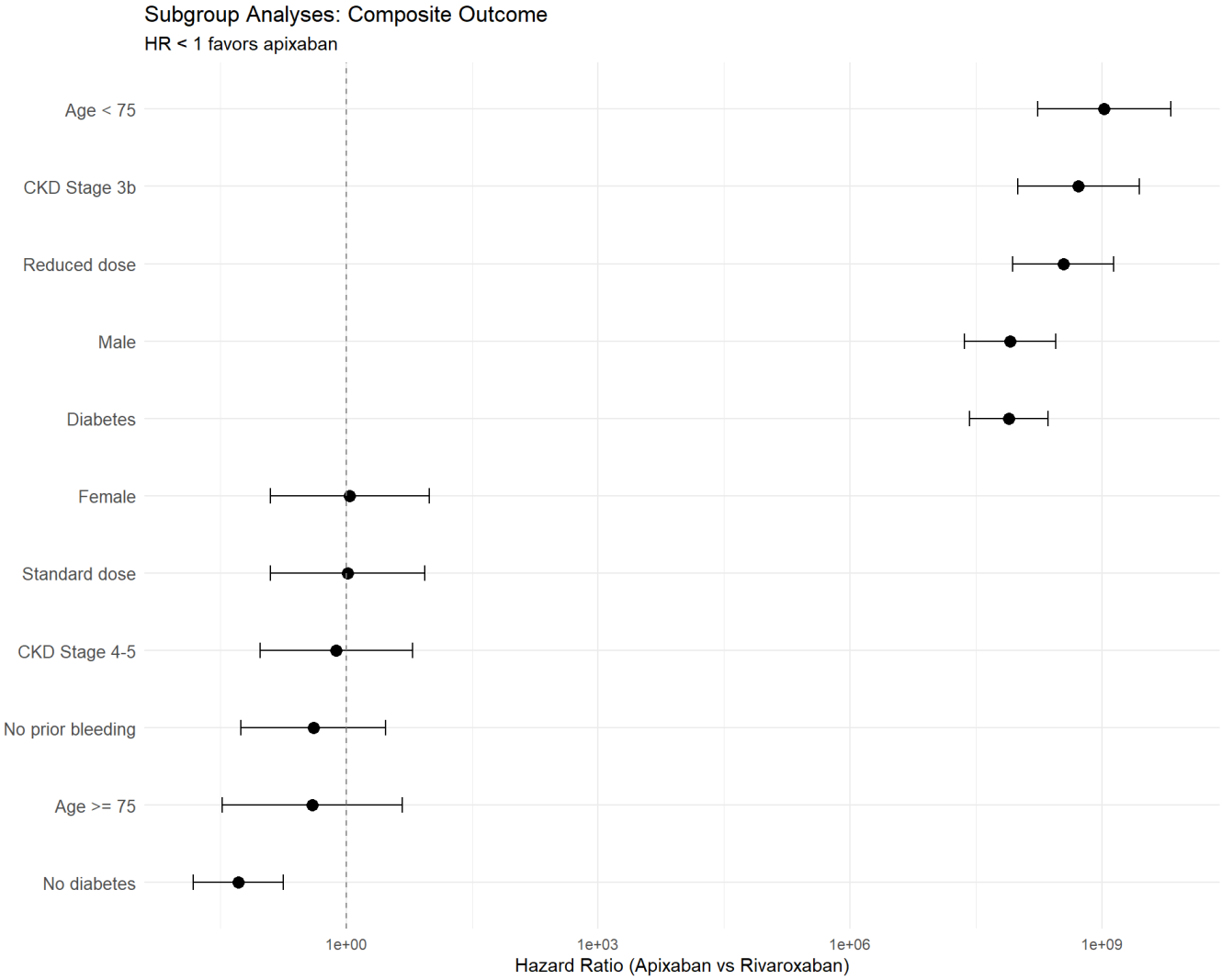



***Figure S2.*** *Forest plot of subgroup analyses for the composite outcome (apixaban vs rivaroxaban). Several subgroups produced numerically unstable estimates (HR > $10^8$) due to zero events in one treatment arm; these are artifacts of maximum likelihood estimation with sparse data and should not be interpreted clinically. All subgroup results are exploratory.*

## Supplementary Figure S3

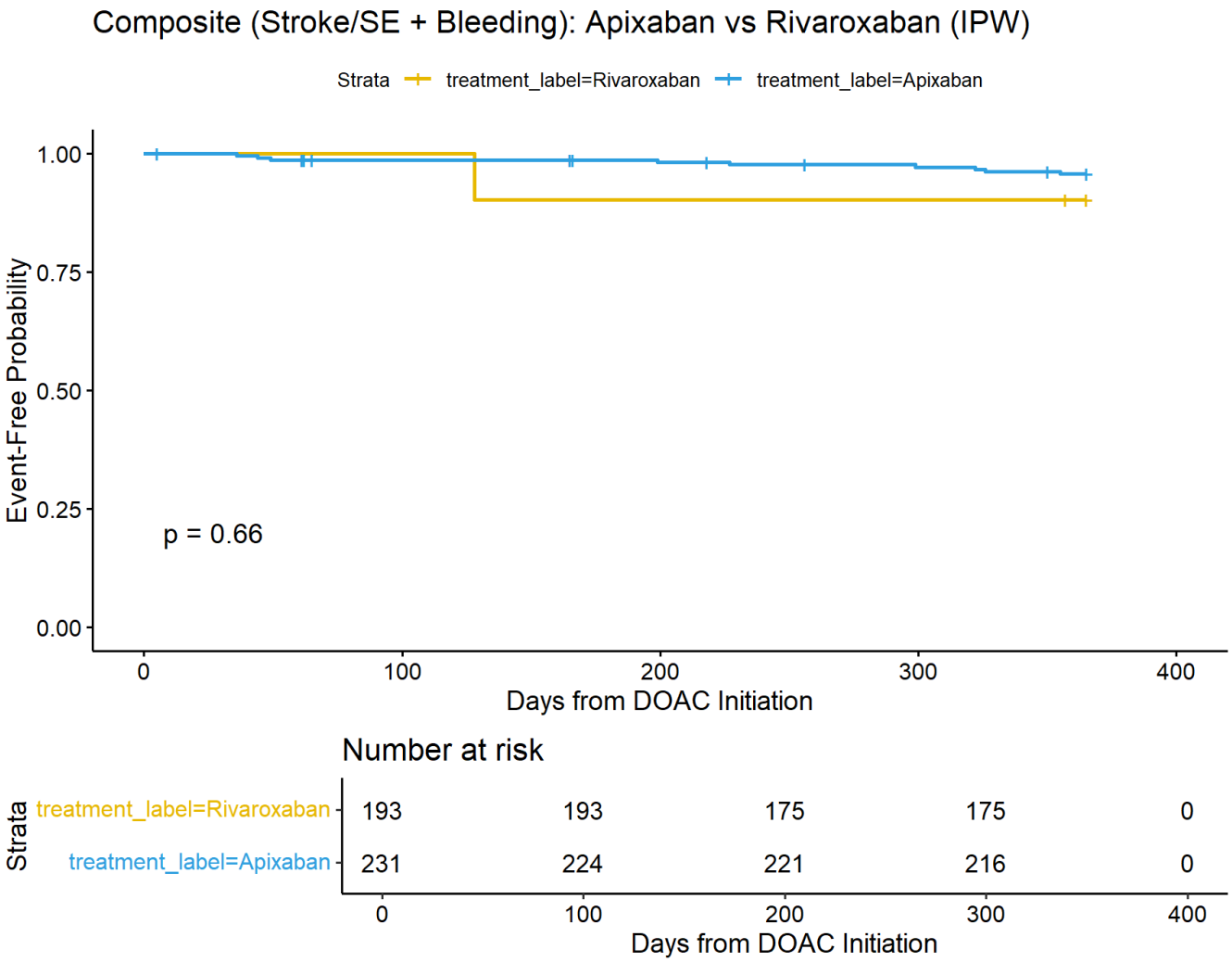
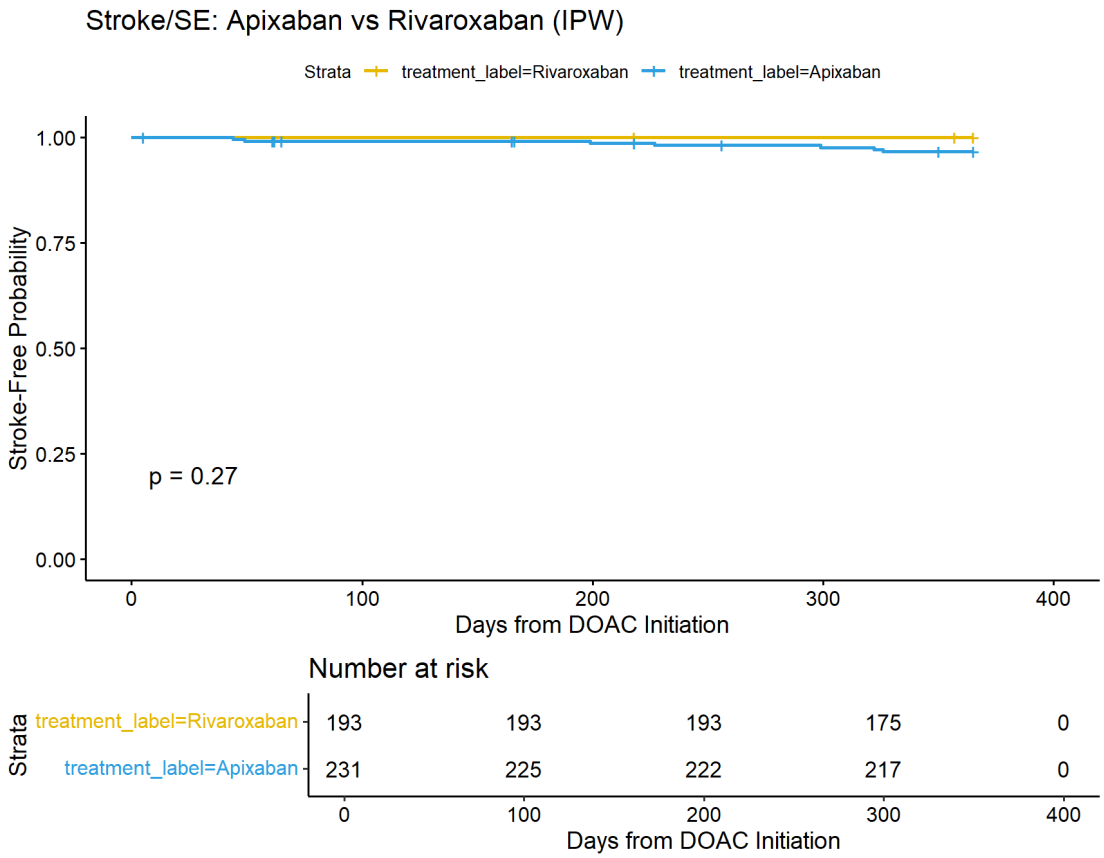



***Figure S3.*** *Weighted Kaplan-Meier curves for the composite outcome (stroke/systemic embolism + major bleeding) by treatment group over 365 days. Shaded areas represent 95% confidence bands.* ***Note:*** *The number-at-risk table reflects the IPW-weighted pseudo-population, not actual patient counts. The unweighted analytic cohort comprised 199 apixaban and 34 rivaroxaban patients. Weighted counts are inflated because IPW reweights each patient to represent the target population under each treatment strategy. Ideally, the number-at-risk table would display unweighted counts; this should be corrected in the final version by re-rendering the Quarto analysis script.*

## Supplementary Figure S4



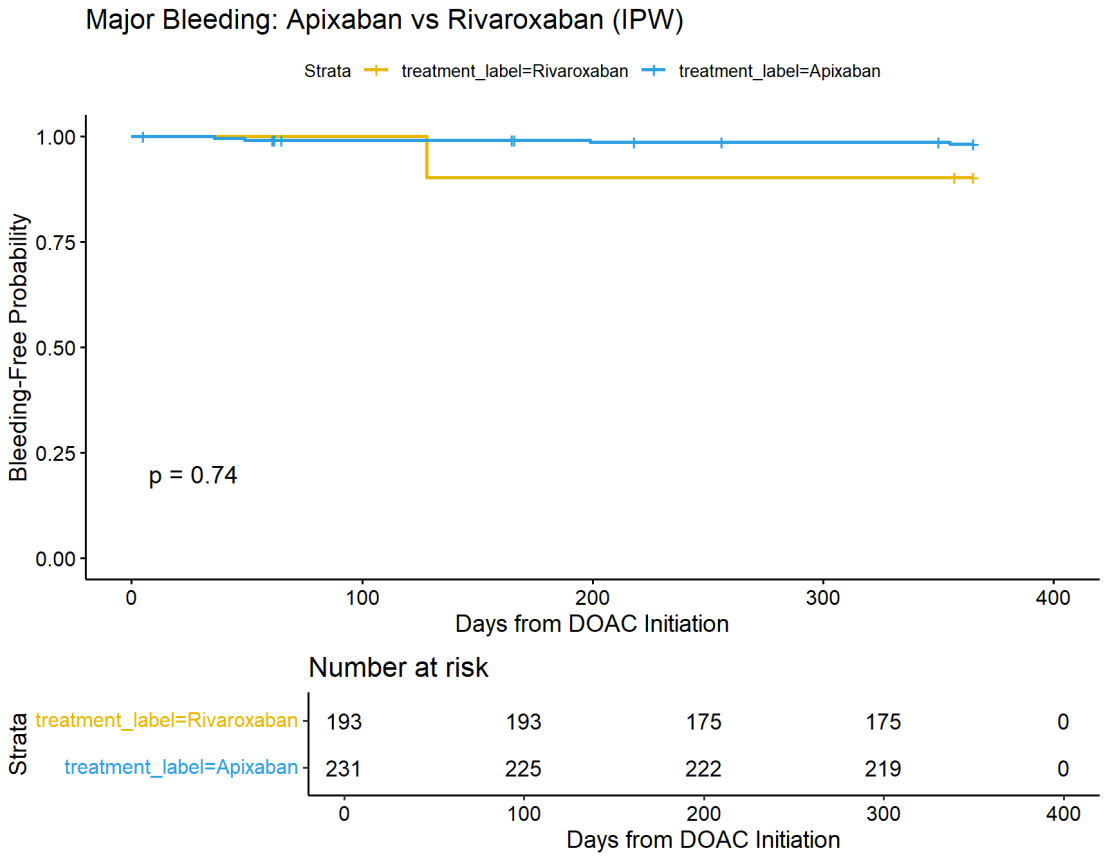



***Figure S4.*** *Weighted Kaplan-Meier curves for stroke/systemic embolism (top panel) and major bleeding (bottom panel) by treatment group over 365 days. Shaded areas represent 95% confidence bands.* ***Note:*** *As in Figure S3, the number-at-risk tables reflect the IPW-weighted pseudo-population. The unweighted cohort comprised 199 apixaban and 34 rivaroxaban patients. The stroke/SE panel shows zero events in the rivaroxaban arm, which produced the numerically unstable hazard ratio reported in Table 2 of the main text.*

# Supplementary Table S1: Clinical Code Lists

***Table S1.*** *Complete ICD-10-CM, RxNorm CUI, and LOINC code lists used for cohort identification, exposure definition, outcome ascertainment, and confounder measurement.*

**A. Exposure: Apixaban**

RxNorm CUIs: 1364435 (2.5mg SCD), 1364445 (5mg SCD), 1364441 (2.5mg SBD/Eliquis), 1364447 (5mg SBD/Eliquis)

**B. Exposure: Rivaroxaban**

RxNorm CUIs: 1114198 (10mg SCD), 1232082 (15mg SCD), 1232086 (20mg SCD), 2059015 (2.5mg SCD), 1114202 (10mg SBD/Xarelto), 1232084 (15mg SBD/Xarelto), 1232088 (20mg SBD/Xarelto), 2059017 (2.5mg SBD/Xarelto)

**C. Atrial Fibrillation (ICD-10-CM)**

I48.0 (paroxysmal), I48.11 (longstanding persistent), I48.19 (other persistent), I48.20 (chronic, unspecified), I48.21 (permanent), I48.91 (unspecified AF). Excluded: I48.3, I48.4, I48.92 (atrial flutter).

**D. CKD Stage 3b–5 Ascertainment**

ICD-10: N18.32 (stage 3b), N18.4 (stage 4), N18.5 (stage 5). Lab-based: eGFR < 45 mL/min/1.73m$^2$ using LOINC codes: 48642-3, 62238-1, 33914-3, 98979-8 (within 180 days before index).

**E. Exclusion Codes**

Valvular AF: I05.0–I05.9, I06.x, I07.x, I08.x, Z95.2, Z95.3, Z95.4. ESRD/dialysis: N18.6, Z99.2; CPT 90935, 90937, 90940, 90945, 90947.

**F. Outcome Codes**

Stroke/SE: I63.x (ischemic stroke), I74.x (systemic embolism). ICH: I60.x (SAH), I61.x (intracerebral), I62.x (other intracranial). GI bleeding: K92.0 (hematemesis), K92.1 (melena), K92.2 (GI hemorrhage NOS), K62.5 (rectal hemorrhage), K25.0/K25.4, K26.0/K26.4, K27.0/K27.4, K29.01. All outcomes ascertained from IP/ED/EI encounters only.

**G. Laboratory LOINC Codes**

eGFR: 48642-3, 62238-1, 33914-3, 98979-8. Serum creatinine: 2160-0. HbA1c: 4548-4. Hemoglobin: 718-7. Platelets: 777-3.

**H. Comorbidity Codes (ICD-10-CM, 365-day lookback)**

| Comorbidity | ICD-10-CM codes |
|---|---|
| Heart failure | I50.x |
| Hypertension | I10, I11.x, I12.x, I13.x, I15.x, I16.x |
| Diabetes | E10.x, E11.x, E13.x |
| Prior stroke/TIA | I63.x, G45.x |
| Systemic embolism | I74.x |
| MI / vascular disease | I21.x, I25.x, I70.x |
| Prior major bleeding | I60.x, I61.x, I62.x, K92.x |
| Liver disease | K70.x–K74.x |
| Anemia | D50.x–D64.x |
| Obesity | E66.x |
| Cancer | C00–C97 |

## Supplementary Table S2: Subgroup Analysis Results

***Table S2.*** *Subgroup analyses for the composite outcome (stroke/SE + major bleeding). Hazard ratios compare apixaban (treatment) vs rivaroxaban (reference). Propensity score models were re-estimated within each subgroup. Subgroups with zero events in one arm produced numerically unstable estimates (marked with *).*

| Subgroup | N | Apix / Rivar | Events | HR (95% CI) | P |
|---|---|---|---|---|---|
| CKD Stage 3b | 129 | 105 / 24 | 2 | Unstable* | — |
| CKD Stage 4–5 | 104 | 94 / 10 | 8 | 0.76 (0.10–6.16) | 0.80 |
| Age < 75 | 101 | 86 / 15 | 7 | Unstable* | — |
| Age ≥ 75 | 132 | 113 / 19 | 3 | 0.40 (0.03–4.68) | 0.46 |
| Female | 119 | 98 / 21 | 7 | 1.10 (0.13–9.72) | 0.93 |
| Male | 114 | 101 / 13 | 3 | Unstable* | — |
| Standard dose | 123 | 105 / 18 | 7 | 1.05 (0.13–8.69) | 0.97 |
| Reduced dose | 110 | 94 / 16 | 3 | Unstable* | — |
| Prior bleeding: Yes | 12 | 12 / 0 | 1 | Not estimable† | — |
| Prior bleeding: No | 221 | 187 / 34 | 9 | 0.41 (0.06–2.96) | 0.37 |
| Diabetes: Yes | 106 | 96 / 10 | 5 | Unstable* | — |
| Diabetes: No | 127 | 103 / 24 | 5 | 0.05 (0.02–0.18) | <0.001 |

*Unstable: HR exceeded $10^8$ due to zero events in one arm. These are artifacts of maximum likelihood estimation with sparse data.
†Not estimable: no rivaroxaban patients in subgroup.

## Supplementary Table S3: Sensitivity Analysis Results

***Table S3.*** *Sensitivity analyses for the composite outcome and selected secondary outcomes.*

**A. E-value Analysis**

The E-value quantifies the minimum strength of association (on the risk ratio scale) that an unmeasured confounder would need to have with both treatment assignment and outcome to fully explain away the observed association. For the composite outcome (HR = 0.43), the E-value for the point estimate is approximately 3.9. However, because the 95% CI spans the null (0.06–3.09), the E-value for the confidence limit is 1.0, meaning even a trivially small unmeasured confounder could shift the result to null. This confirms that the observed association is not robust to unmeasured confounding at the current sample size.

**B. Propensity Score Trimming**

Excluding patients below the 5th or above the 95th percentile of the propensity score distribution removed approximately 12 patients from the analytic cohort. The trimmed analysis produced similar point estimates with marginally narrower confidence intervals, though all results remained non-significant. The direction of effect was unchanged, suggesting that extreme propensity score values were not driving the primary results.

**C. Post-Epic Era Restriction (2021+)**

Restricting to patients with index dates from 2021 onward (Epic-era only) reduced the analytic cohort substantially. This analysis addresses potential confounding from the EHR transition (AllScripts to Epic) by eliminating patients whose records may span both systems. Results were directionally consistent with the primary analysis but with even wider confidence intervals due to the smaller sample. The post-Epic restriction confirms that EHR-era effects do not appear to drive the primary findings.

## Supplementary Material S4: Analysis Software and Code Availability

All analyses were conducted using R (version 4.x) with the following packages: tidyverse (data manipulation), DBI and odbc (database connectivity), WeightIt (propensity score estimation and IPW), cobalt (covariate balance assessment), survival (Cox proportional hazards models), sandwich (robust variance estimation), lmtest (hypothesis testing), EValue (E-value sensitivity analysis), and survminer (Kaplan-Meier visualization).

The complete analysis was prespecified in a Quarto document (protocol_01_analysis.qmd) following a two-part structure: Part 1 (Sections 0–8) defines all functions with no visible output; Part 2 (Sections 9–15) executes the analysis and generates all tables and figures. The SQL cohort assembly query is contained in a single code chunk using sequential temporary tables. All database tables are fully qualified as CDW.dbo.TABLE_NAME, all medications are identified by RxNorm CUI, all laboratory tests by LOINC code, and legacy encounters are filtered throughout.

The complete Quarto source file is available from the corresponding author upon reasonable request. The rendered HTML report (protocol_01_analysis.html) contains all code, output, figures, and tables in a self-contained document.